

Derivation for move_position_by_delta_on_axis in src/camera.rs

Abstract. The function `move_position_by_delta_on_axis` is responsible for translating the camera about it's target, transforming a linear distance along an axis into a circular motion along the same axis. As such, the derivation for the function is important to the accuracy and reproducibility of its processes.

1 Function Definitions

The function in question is:

```
move_position_by_delta_on_axis(
    &self, axis:CameraAxis, delta:f32
) -> Result<Vector<f32>, MatrixError>
```

The function takes inputs of:

1. `&self`, of type `Self`, which is type `CameraInfoMatrix`
2. `axis:CameraAxis`, which has enum variants of `Forward`, `Right`, and `Up`
3. `delta:f32`, which is the distance `delta` to move along the specified axis

The function produces an output of either:

1. `Ok(Vector<f32>)`, which is the vector that gets added to the current position to translate it to the new position
2. `Err(MatrixError)`, signalling that a math error occurred

2 Initial Process

To create an initial mechanism for producing the desired translation vector, the following process is observed:

1. The specified axis is matched with the corresponding vector camera axis
 - 1.1 important note: as movement along the forwards axis is governed by a more simple process, passing an axis of `CameraAxis::Forwards` will return an error, to prevent incorrect usage.

2. Let the unit vector representing the camera axis be $\vec{a}$
3. Define the current camera position and camera target as $\vec{P}$ and $\vec{T}$, respectively
4. The vector describing the direction that is going from the target towards the camera is:

$$\vec{V} = \vec{P} - \vec{T} \quad (1)$$

- 4.1 By considering a circle whose centre is the target, and wherein the camera position lies along its circumference, it can be found that the radius of this circle is the distance between the camera and it's target:

$$r = \|\vec{V}\| = \|\vec{P} - \vec{T}\| \quad (2)$$

- 4.2 It is also to be noted that this radius will remain constant after the transformation, as the camera will be moved around the target in a circle with radius r
5. Define the scalar distance that the camera will be translated to be Δ
 - 5.1 The vector describing the change in position along the axis $\vec{a}$ by an amount Δ is:

$$\vec{\Delta} = \Delta \vec{a} \quad (3)$$

6. By adding the change $\vec{\Delta}$ to the current camera direction $\vec{V}$, we get the direction altered by the distance Δ along the axis $\vec{a}$, where:

$$\vec{V}'_i = \vec{V} + \vec{\Delta} = \vec{P} - \vec{T} + \vec{\Delta} \quad (4)$$

- 6.1 Note that the distance $\|\vec{V}'_i\| \neq \|\vec{V}\|$. As such the distance between the new position and the target no longer equals r , meaning that the camera is no longer along the desired circle of radius r .

7. By normalising the new camera vector, the new direction of the camera vector $\vec{V}_{i_n}'$ is obtained, where

$$\vec{V}_{i_n}' = \frac{1}{\|\vec{V}_i'\|} \vec{V}_i' \quad (5)$$

- 7.1 To place the camera vector onto the initially desired radius, all that is required is to multiply it by said radius. As such:

$$\vec{V}_{i_r}' = r \vec{V}_{i_n}' \quad (6)$$

8. Finally, to obtain the vector representing the new position of the camera in space, the target vector $\vec{T}$ is taken and added to the new direction vector, to obtain the final position after the transformation, so that the final position $\vec{P}'$ is

$$\vec{P}' = \vec{T} + \vec{V}_{i_r}' \quad (7)$$

To further progress on the initial process, and render it more useful and intuitive, it is first beneficial to render final formula down to its base terms.

$$\begin{aligned} \vec{P}' &= \vec{T} + \vec{V}_{i_r}' \\ &= \vec{T} + r \vec{V}_{i_n}' \\ &= \vec{T} + \|\vec{P} - \vec{T}\| \frac{1}{\|\vec{V}_i'\|} \vec{V}_i' \\ &= \vec{T} + \frac{\|\vec{P} - \vec{T}\|}{\|\vec{V} + \vec{\Delta}\|} (\vec{V} + \vec{\Delta}) \\ &= \vec{T} + \frac{\|\vec{P} - \vec{T}\|}{\|\vec{P} - \vec{T} + \Delta\vec{a}\|} (\vec{P} - \vec{T} + \Delta\vec{a}) \end{aligned} \quad (8)$$

Now, given that the initial camera position $\vec{P}$ becomes the new position $\vec{P}'$, there must exist some vector $\vec{X}$ such that $\vec{P} + \vec{X} = \vec{P}'$. Otherwise stated:

$$\begin{aligned} \vec{X} &= \vec{P}' - \vec{P} \\ &= \vec{T} + \frac{\|\vec{P} - \vec{T}\|}{\|\vec{P} - \vec{T} + \Delta\vec{a}\|} (\vec{P} - \vec{T} + \Delta\vec{a}) - \vec{P} \\ &= \frac{\|\vec{P} - \vec{T}\|}{\|\vec{P} - \vec{T} + \Delta\vec{a}\|} (\vec{P} - \vec{T} + \Delta\vec{a}) - (\vec{P} - \vec{T}) \\ &= \frac{\|\vec{V}\|}{\|\vec{V} + \Delta\vec{a}\|} (\vec{V} + \Delta\vec{a}) - \vec{V} \end{aligned} \quad (9)$$